



Thinking About Domain and Range

Sometimes, only some of the domain and range values of a function make sense for a given situation.

Two Types of Domains and Ranges

Definitions

The **theoretical domain and range** of a function are the sets of all allowable inputs and outputs. The **practical domain and range** of a function are the sets of all realistic inputs and outputs for a particular situation.

EXAMPLE 1

A vendor has 20 umbrellas to sell. He sells them for \$15 each. The function $m(u) = 15u$ models the total amount of money the vendor makes from selling u umbrellas.

A What is the theoretical domain of the function?

SOLUTION

You can substitute any real number into the function rule, $15u$. The theoretical domain is the set of all real numbers.

B What is the practical domain of the function?

SOLUTION

The fewest number of umbrellas the vendor can sell is 0. The greatest number of umbrellas the vendor can sell is 20. It does not make sense for the vendor to sell part of an umbrella, so the practical domain is the set of all integers from 0 to 20, inclusive.

C What is the theoretical range of the function?

SOLUTION

Any real number could be the result of multiplying another real number by 15. The theoretical range is the set of all real numbers.

D What is the practical range of the function?

SOLUTION

The total amount of money the vendor makes is \$15 times the number of umbrellas he sells. He can sell as few as 0 and as many as 20 umbrellas, so he can make as little as \$0 and as much as \$300. The practical range is the set of all multiples of 15 between 0 and 300, inclusive. ■

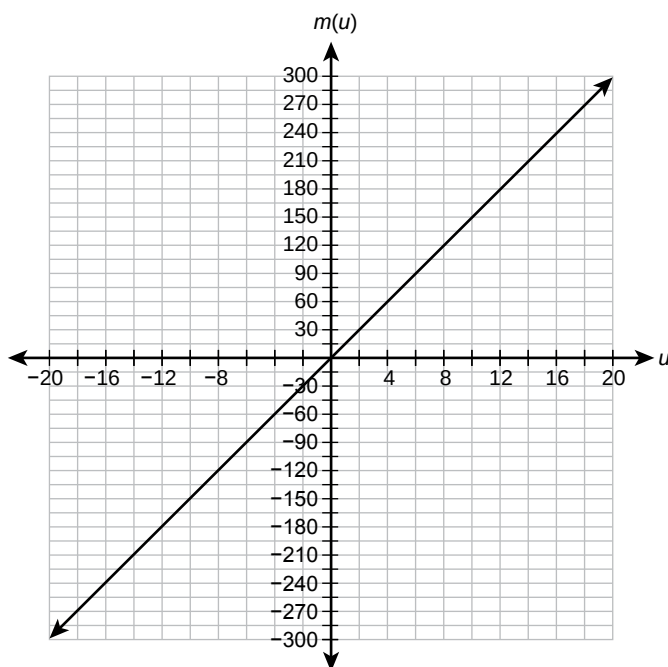
► **Think About It** You can write the practical range as $\{m(u) \mid m(u) = 15k, k \in \mathbb{Z}, k \geq 0, k \leq 20\}$.

EXAMPLE 2

Graph $m(u)$ from Example 1 over its theoretical domain and over its practical domain.

A What is the theoretical domain of the function?

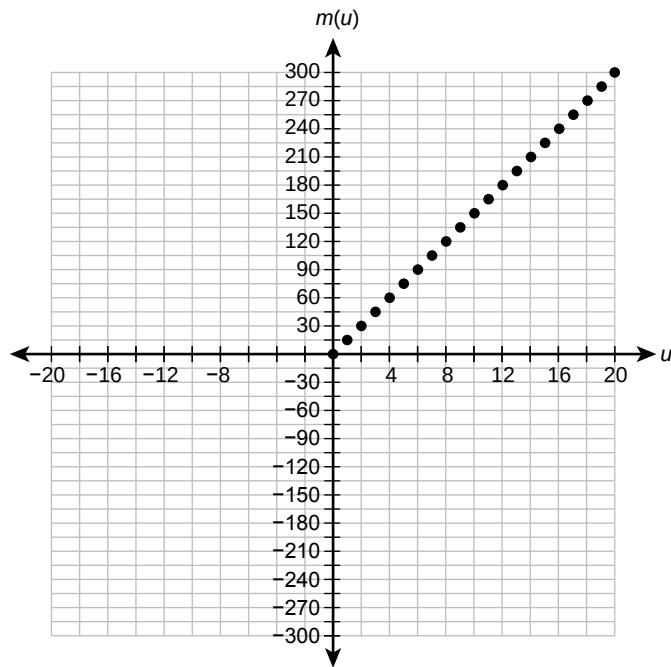
SOLUTION



theoretical domain: $\{u \mid u \in \mathbb{R}\}$

B What is the practical domain of the function?

SOLUTION



practical domain: $\{u \mid u \in \mathbb{Z}, u \geq 0, u \leq 20\}$ ■